EXECUTIVE BRIEF & EXPEDITION ARCHITECTURE

Humanity's First Permanent Interplanetary Settlement Protocol

MISSION MANDATE

Mission: Mars is humanity's landmark enterprise to establish an enduring, self-sustaining presence on the Red Planet. Chronicling the complete journey from Earth orbital launch, through 54.6 million kilometers of deep space transit, to touchdown and the operational establishment of Ares Base Alpha.

MISSION STAGES & OPERATIONAL SEQUENCE

PHASE	NAME	OPERATIONAL PROFILE
STAGE 01	EARTH	The Cradle of Humanity
STAGE 02	LAUNCH	Trans-Mars Injection
STAGE 03	DEEP SPACE	Hohmann Transfer
STAGE 04	MARS ARRIVAL	Atmospheric Entry
STAGE 05	EXPLORATION	Surface Reconnaissance
STAGE 06	COLONY	Ares Base Alpha
STAGE 07	THE FUTURE	Interplanetary Species

CRITICAL PLANETARY CONSTANTS (EARTH vs MARS)

PARAMETER	EARTH BASELINE	MARS TARGET	ENGINEERING IMPLICATION
Surface Gravity	9.81 m/s² (1.00 g)	3.72 m/s² (0.38 g)	Demands strict resistance exercise
Atmospheric Pressure	101.3 kPa (1.00 atm)	0.61 kPa (<1%)	Armstrong limit: Blood boils without suit
Mean Temperature	+14.9 °C	" 6 3 . 0 ° C	S w i n g s f r o m " 1 4 0 ° C t o + 2 0 ° C
Dominant Gas	78% Nitrogen / 21% O2	95.3% CO2	Feedstock for MOXIE oxygen generation
Annual Radiation	0.33 mSv / year	240 – 300 mSv / yr	Requires 3m regolith shielding
Radio Signal Lag	< 0.05 seconds	3.1 to 22.3 minutes	Mandates total vehicle autonomy

SPACECRAFT SUBSYSTEMS & TRANSIT TELEMETRY

Ares Cycler Interplanetary Vehicle Specifications

PRIMARY FLIGHT SUBSYSTEMS

PROPULSION ARRAY

NOMINAL

High-thrust methane and liquid oxygen engines execute the initial Trans-Mars Injection burn, followed by low-thrust continuous Hall-effect ion propulsion to fine-tune trajectory.

Thrust (Vacuum): **6,600 kN** Specific Impulse (Isp): **382 s / 3,400 s (Ion)** Propellant Mass: **1,200 Metric Tons** TMI Delta-V: **3,800 m/s**

LIFE SUPPORT (ECLSS)

NOMINAL

Advanced solid amine vacuum desorption loops scrub metabolic carbon dioxide, while upgraded Sabatier reactors synthesize potable water and breathable oxygen.

Oxygen Recovery: **98.4 %** Water Recycling: **98.8 %** Internal Pressure: **101.3 kPa (1 atm)** CO2 Removal Rate: **1.2 kg / crew / day**

DEEP SPACE COMM ARRAY

ACTIVE

Laser communications revolutionize deep space exploration, sending high-definition scientific data and 4K telepresence feeds across hundreds of millions of kilometers.

Laser Downlink: **250 Mbps** RF Backup Link: **12 Mbps** Current Latency: **14 min 22 sec** Pointing Accuracy: **0.00008 deg**

AUTONOMOUS NAVIGATION

OPTIMIZED

Because round-trip radio signals take tens of minutes, the spacecraft calculates its trajectory, gravitational perturbations, and trim maneuvers completely autonomously.

Star Tracker FOV: **28 deg (Dual Head)** Position Accuracy: **± 12.4 m** Guidance Refresh: **100 Hz** Autonomy Level: **Level 5 (Unattended)**

TRAJECTORY & LAUNCH WINDOW LOGISTICS (HOHMANN MINIMUM ENERGY)

Earth and Mars reach orbital opposition every 26 months (780 days). Trans-Mars Injection burns require delta-V ~3.8 km/s from Low Earth Orbit. Transit duration ranges between 180 and 210 days. Planetary approach requires either high-energy propulsive retro-burns or guided hypersonic aerobraking through the Martian upper mesosphere.

SURFACE TOPOGRAPHY & RECONNAISSANCE SITES

Geological Targets, Ancient River Deltas & Water Ice Reservoirs

PRIMARY GEOGRAPHIC & SCIENTIFIC LANDMARKS

OLYMPUS MONS — Tharsis Volcanic Region (Shield Volcano)

COORDS: 18.65° N, 226.2° E | DATUM: 21.9 km (72,000 ft) | SCALE: 624 km

The tallest planetary mountain in the Solar System. Its peak extends almost three times higher than Mount Everest, penetrating past the dense lower atmosphere into near-vacuum.

EXPLORATION MANDATE: Ancient lava tube networks along the caldera slopes provide natural subterranean radiation shielding for

VALLES MARINERIS — Equatorial Tectonic Trough (Tectonic Canyon System)

COORDS: 13.9° S, 300.8° E | DATUM: " 7.0 km below datum | SCALE: 4,000 km length

A colossal tectonic chasm spanning roughly one-fifth of the circumference of Mars. If placed on Earth, it would stretch from New York to Los Angeles.

EXPLORATION MANDATE: Exposes deep crustal geological stratigraphy spanning four billion years of Martian history, with recurrent

JEZERO CRATER — Syrtis Major Quadrangle (Paleo-Lake Delta)

COORDS: 18.38° N, 77.58° E | DATUM: " 2.5 km | SCALE: 45 km

The historic touchdown zone of the NASA Perseverance rover. A billion years ago, a prominent river breached the crater rim, depositing an immense sediment delta.

EXPLORATION MANDATE: Fine-grained clay minerals preserve prime conditions for the fossilization of microscopic organic

PLANUM BOREUM — Northern Polar Plateau (Perennial Ice Cap)

COORDS: 88.0° N, 0.0° E | DATUM: 2.0 km above plains | SCALE: 1,000 km

The vast northern polar cap, composed primarily of water-ice layered with atmospheric dust, capped by a seasonal winter mantle of solid carbon dioxide frost.

EXPLORATION MANDATE: Contains an estimated 1.6 million cubic kilometers of water ice—enough to sustain a human colony for

ARCADIA PLANITIA — Lowland Volcanic Plain (Ice-Rich Alluvial Plain)

COORDS: 39.3° N, 189.7° E | DATUM: " 4.0 km | SCALE: 1,800 km

A smooth, geologically stable volcanic plain characterized by shallow, accessible sheets of clean glacial water ice buried just 1 to 2 meters beneath the surface regolith.

EXPLORATION MANDATE: Prime landing and settlement candidate due to low elevation (higher atmospheric density aids braking)

ARES BASE ALPHA & THE INTERPLANETARY FUTURE

Permanent Habitat Infrastructure, ISRU Propellant & Epilogue

SETTLEMENT ARCHITECTURE SPECIFICATIONS (ARES BASE ALPHA)

PRESSURIZED GEODESIC HABITAT [CAPACITY: 32 Astronauts] | [Crew Living Quarters] | POWER: 45 kW | SHIELDING: 3.0 m Compacted Regolith Berm

Tri-level interlocking subterranean vaults providing crew quarters, medical infirmary, galley, recreational space, and dual redundant airlocks.

CORE FUNCTION: Provides habitable sanctuary and radiation protection for all long-duration expedition personnel.

AEROPONIC AGRICULTURAL DOME [CAPACITY: Supports 4 Crew Members] | [Primary Biomass Production] | POWER: 85 kW (LED PAR Spectrum) | SHIELDING: ETFE Double-Glazed UV-C Filter

High-density vertical aeroponic farming bays cultivating dwarf wheat, potatoes, spirulina algae, soybeans, and nitrogen-fixing legumes under

CORE FUNCTION: Ensures dietary independence from Earth supply chains and closes the biological oxygen-carbon loop.

ASTROBIOLOGY & GEOLOGY LAB [Scientific Analysis] | [CAPACITY: 12 Researchers] | POWER: 60 kW | SHIELDING: Reinforced Sintered Shell

ISO-4 cleanrooms equipped with automated electron microscopy, DNA sequencer chips, mass spectrometers, and sample quarantine

CORE FUNCTION: Unlocks the answers to whether life ever emerged independently on Mars.

NUCLEAR & SOLAR ENERGY HUB [CAPACITY: Autonomous Remote Operation] | [Primary Power Supply: 1.2 MW Continuous] | SHIELDING: Subsurface Placement / Borated Concrete

A resilient hybrid grid combining four 10-kWe Kilopower Stirling fission reactors buried in remote craters with a 10-hectare dust-repellent

CORE FUNCTION: Supplies uninterrupted megawatt-class electricity to life support, ISRU chemical plants, and rovers.

PRESSURIZED ROVER & LOGISTICS BAY [CAPACITY: Maintains 2 Pressurized Rovers + 4 Crew] | [Surface Mobility & EVA Hub] | POWER: 50 kW | SHIELDING: Regolith Arched Hangar

A pressurized maintenance depot capable of docking multi-crew long-range pressurized rovers, servicing robotic haulers, and staging 14-day

CORE FUNCTION: Expands the expedition exploration radius up to 500 kilometers from the central outpost.

ISRU FUEL PLANT & RETURN PAD [CAPACITY: Autonomous Propellant Synthesis] | [Return Launch Operations] | POWER: 450 kW (Peak Electrolysis) | SHIELDING: Blast Deflector Berms

The In-Situ Resource Utilization plant mines subsurface water ice and captures atmospheric carbon dioxide, converting them via the

CORE FUNCTION: Enables the return leg of the mission, proving the feasibility of reusable interplanetary flight.

THE INTERPLANETARY IMPERATIVE

"THE FUTURE IS NOT JUST SOMEWHERE WE GO. IT IS SOMETHING WE BUILD."

Ares Odyssey Expedition 01 · Certified by International Mars Exploration Directorate